# Gate-first F6-IC on x86 — run report

Run `run-20261016T2232Z` · binary `f6_ic_probe` built from `f25d5e520` (origin/main) · 2026-10-16

## What ran

`scripts/run_arms.sh`: `f6_ic_probe --engines f4,f6p --reps 3` (`f4` inherited F4, `f6p` F6-IC with the shared pair-sum index), one binary, arms set by environment only — **off** (no switch), **gate** (`KIC_F6_GATE_FIRST=1`), **rref** (`KIC_F4_INHERIT_ROOT=rref`), **gaterref** (both) — on nine cells: the four of `research/lit2026_w5_lane_batched_f4_20261011` (K1/2^17 d6, K1/2^23 d7, K0/2^23 d8, K0/2^31 d9) and five new ones (K0/2^19 d7, K1/2^19 d6, K1/2^17 d7, K1/2^29 d8, K0/2^37 d9), m = 3, `max_degree` 3. 5 rounds, arm order rotated each round. Host: Intel Xeon @ 2.10 GHz, 4 vCPU KVM container, BMI2 present, so the inherited-F4 eager rewrite (`KIC_F4_EAGER`, default) is in use — the path the aarch64 study did not measure. Shared, unpinned, L0; load (1-min) 1.02–4.21. A repository search ran on one core during round 1 (22:33–22:36 UTC). 180 arm runs requested, 180 exited 0.

Stage unit = the probe's per-arm total `word_xors` (one pass, deterministic). CPU = the probe's `cpu_ms` (process user+system over its three repetitions). Ratios are derived: off / arm of the stage units, and of the medians over the 5 rounds for CPU; the bracket is (min off / max arm) – (max off / min arm) over rounds.

## Exactness

Every one of the 80 targets (cell × engine × target) has the same decision tuple — hit, indices, exhausted, reductions, splits, propagations, infeasible, refutations, witnesses, partial refutations, geometric additions — in every arm and every round. Every arm's stage unit is identical in all 5 rounds.

## F6-IC (`f6p`), gate-first against off

| cell | units off | units gate | off/gate | CPU off ms | CPU gate ms | CPU off/gate |
|---|---|---|---|---|---|---|
| K1_17 | 15514870 | 10830979 | 1.433 | 258.7 | 163.1 | 1.59 (1.37–2.25) |
| K1_23 | 59881009 | 38122485 | 1.571 | 679.1 | 320.7 | 2.12 (1.56–2.27) |
| K0_23 | 132377695 | 81022008 | 1.634 | 1461.7 | 752.8 | 1.94 (1.72–2.41) |
| K0_31 | 285541596 | 164736149 | 1.733 | 2267.1 | 1156.9 | 1.96 (1.85–2.20) |
| K0_19 | 13660081 | 10226773 | 1.336 | 181.8 | 116.5 | 1.56 (1.25–1.75) |
| K1_19 | 28309592 | 19117122 | 1.481 | 400.2 | 206.5 | 1.94 (1.59–2.53) |
| K1_17d7 | 15297452 | 10867029 | 1.408 | 278.0 | 132.3 | 2.10 (1.69–2.63) |
| K1_29 | 182738266 | 112675706 | 1.622 | 1554.7 | 872.4 | 1.78 (1.69–2.06) |
| K0_37 | 644803622 | 376487242 | 1.713 | 3947.2 | 2097.3 | 1.88 (1.60–2.07) |

Gate-first skipped 447 / 432 / 840 / 405 refuted nodes on the four W5 cells, the same counts the aarch64 study recorded.

## Inherited F4 (`f4`), gate-first against off

| cell | units off | units gate | off/gate | CPU off ms | CPU gate ms | CPU off/gate |
|---|---|---|---|---|---|---|
| K1_17 | 18019804 | 18019804 | 1.000 | 474.0 | 468.7 | 1.01 (0.78–1.20) |
| K1_23 | 62213299 | 62213299 | 1.000 | 776.8 | 842.5 | 0.92 (0.81–1.08) |
| K0_23 | 167017420 | 167017420 | 1.000 | 2487.9 | 2425.8 | 1.03 (0.98–1.12) |
| K0_31 | 290412310 | 290412310 | 1.000 | 2367.7 | 2439.6 | 0.97 (0.85–1.09) |
| K0_19 | 17250812 | 17250812 | 1.000 | 336.5 | 338.9 | 0.99 (0.76–1.21) |
| K1_19 | 30159678 | 30159678 | 1.000 | 587.6 | 567.4 | 1.04 (0.92–1.34) |
| K1_17d7 | 22448248 | 22448248 | 1.000 | 579.1 | 491.3 | 1.18 (0.75–1.35) |
| K1_29 | 185919270 | 185919270 | 1.000 | 1713.3 | 1747.2 | 0.98 (0.90–1.13) |
| K0_37 | 652414757 | 652414757 | 1.000 | 4194.7 | 4128.3 | 1.02 (0.93–1.15) |

## F6-IC, the RREF root (`KIC_F4_INHERIT_ROOT=rref`)

| cell | units off/rref | units off/gaterref | CPU off/rref | CPU off/gaterref |
|---|---|---|---|---|
| K1_17 | 0.931 | 1.275 | 0.90 | 1.73 |
| K1_23 | 1.005 | 1.488 | 1.12 | 2.33 |
| K0_23 | 1.102 | 1.660 | 1.08 | 2.20 |
| K0_31 | 1.226 | 1.934 | 1.18 | 2.12 |

| | | | | |
|---|---|---|---|---|
| K0_19 | 0.911 | 1.138 | 1.03 | 1.62 |
| K1_19 | 0.976 | 1.324 | 1.01 | 1.77 |
| K1_17d7 | 0.889 | 1.180 | 0.98 | 1.86 |
| K1_29 | 1.101 | 1.637 | 1.13 | 1.88 |
| K0_37 | 1.227 | 1.911 | 1.17 | 2.04 |

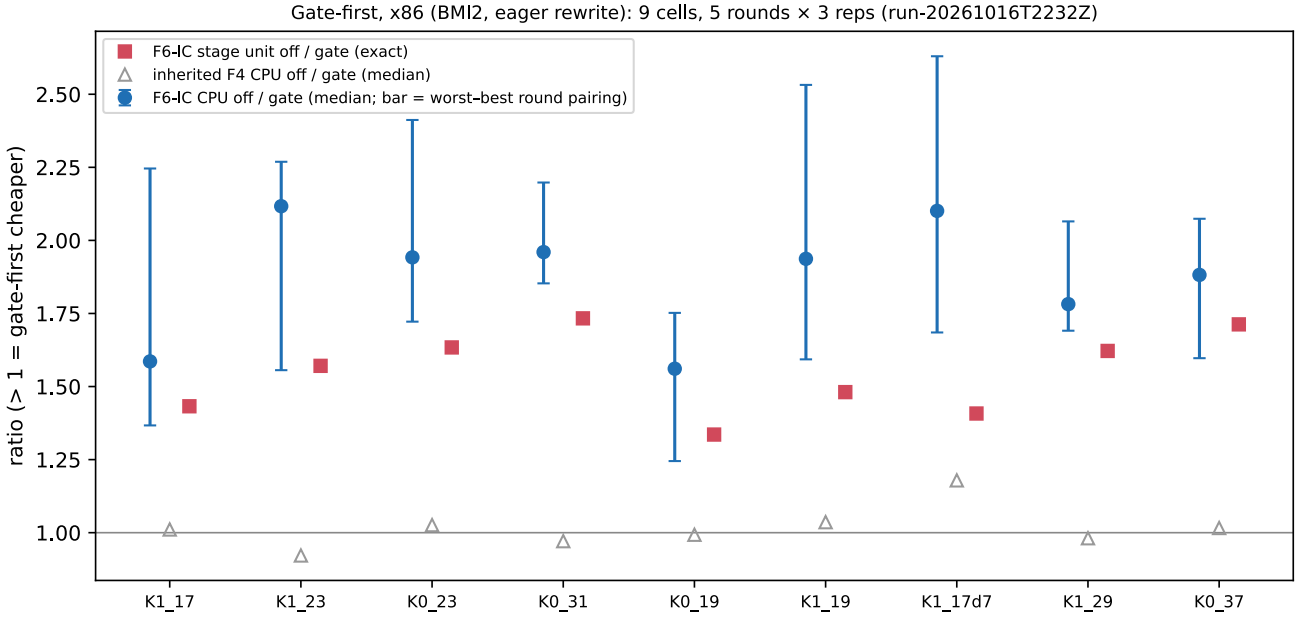



Figure 1: Gate-first ratios per cell, run-20261016T2232Z, 5 rounds.

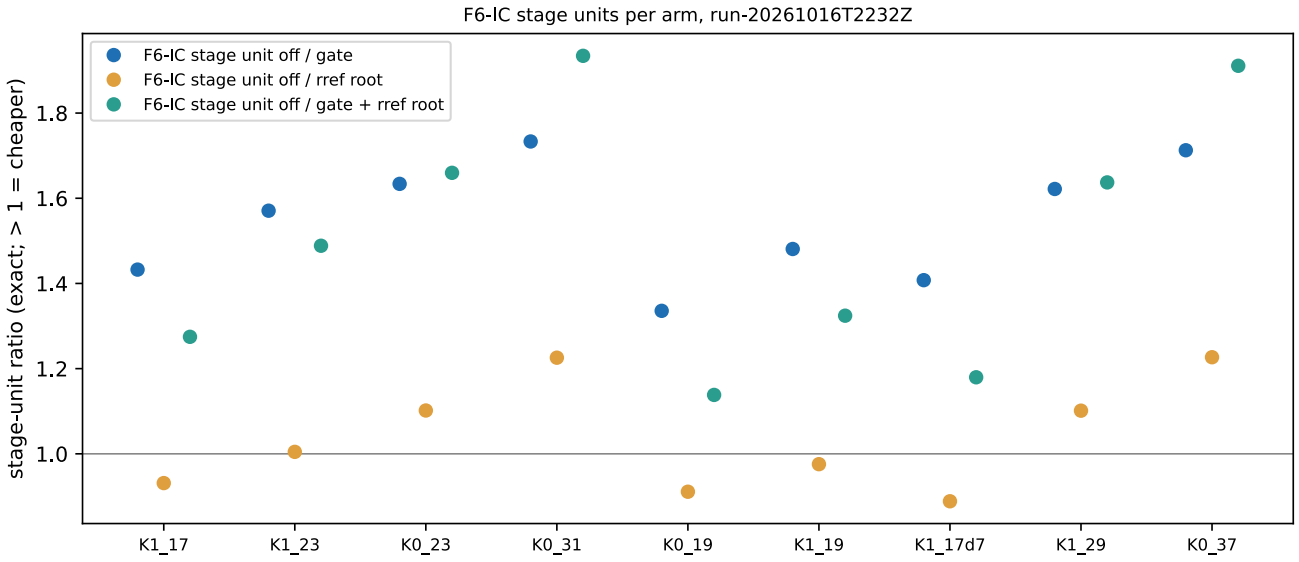



Figure 2: F6-IC stage-unit ratios of the three switched arms, run-20261016T2232Z.

## Observations

- Gate-first, F6-IC: 1.336–1.733× fewer stage units on all nine cells; median CPU 1.56–2.12× lower; the worst round pairing is above 1 on every cell (lowest 1.25, K0/2^19).
- Gate-first, inherited F4: stage units identical (the engine has no node oracle); CPU medians 0.92–1.18×, within this host's noise.
- RREF root: on x86 it lowers stage units on the larger cells (1.10–1.23× on K0/2^23, K0/2^31, K1/2^29, K0/2^37) and raises them on the smaller ones (0.89–0.98× on K1/2^17 d6 and d7, K0/2^19, K1/2^19); K1/2^23 is level. The aarch64 study found it lower on all four of its cells.

## Records

`raw/r<round>-<cell>-<arm>.jsonl` (first line: arm, cell, round, UTC, load), `raw/*.status`, `host.txt`, `summary.txt` and `table.tsv` (from `../../summarize.py`), `fig_*.svg|png` (from `../../plot.py`).